

Part 1: Exploratory Analytics & Information Dashboards

1. Domain Overview:

Selected Domain: Residential Property Market Analytics in Ireland

Domain Description:

The domain selected for this assignment is **Residential Housing Market Analytics**, with a specific focus on **property transaction prices and sales dynamics across Ireland**.

The Irish housing market plays a critical role in:

- Household wealth accumulation
- Urban and regional planning
- Government housing policy
- Real estate investment decisions

In recent years, Ireland has experienced:

- Rapid population growth
- Increased housing demand
- Supply shortages
- Significant regional price disparities

As a result, analysing housing transaction data using **exploratory data analytics and visual dashboards** is essential for understanding market behaviour and supporting evidence-based decision-making.

This domain is particularly suitable for this assignment because:

- It contains **rich temporal data** (daily, monthly, quarterly, yearly)
- It has **strong geographic components** (county-level analysis)
- It enables **comparative, distributional, and trend-based visualisations**
- It is highly relevant to **policy makers and property investors**

2. Data Source & Reliability

Dataset Used: The dataset used for this project is the **Irish Residential Property Price Register (PPR)**.

Data Publisher

- Property Services Regulatory Authority (PSRA)

Accessed Through

- Ireland's official open data portal

Dataset Links

- <https://www.propertypriceregister.ie>

Data Reliability Assessment

The dataset is **highly reliable** because:

- All residential property transactions in Ireland **must be legally reported** to PSRA
- It contains **actual transaction prices**, not estimates or asking prices
- It is **government-verified and publicly audited**
- It is widely used by **economists, planners, and real estate professionals**

Therefore, the dataset provides an **accurate and authoritative representation of Ireland's housing market**.

3. Variable Relationships

Eircode	County where property is located	Geographic comparison
Year	Address	Spatial accuracy
RealPrice (€)	Actual transaction price	Primary measure
PropertyCondition	New or Second-Hand	Market structure
PropertyType	Detached, Apartment, etc.	Price variation
PriceCategory	Low / Medium / High	Market segmentation
HasEircode	Yes / No flag	Data completeness

- **Price** is influenced by:
 - Location (County)
 - PropertyCondition (New vs Second-Hand)
 - PropertyType
- **Time variables** (Month, Quarter, Year) reveal:
 - Seasonal patterns
 - Long-term trends
- **PriceCategory + County** shows:
 - Regional affordability gaps
- **Sales Volume + Time** indicates:
 - Market activity cycles

These relationships form the analytical foundation for all dashboards.

4. Analytical Questions & Issues Explored

The following key questions guided the exploratory analysis:

Pricing Trends

1. How have property prices changed over time in Ireland?
2. Are prices rising uniformly across counties?

Geographical Differences

3. Which counties are the most and least expensive?
4. Are counties near Dublin more expensive?

Market Structure

5. Are new properties consistently priced higher than second-hand?
6. What is the distribution of low, medium, and high-priced properties?

Sales Dynamics

7. When does sales activity peak during the year?
8. Are there seasonal fluctuations in prices and volumes?

Affordability & Volatility

9. Which counties show the highest price volatility?
10. Where is affordability most constrained?

5. Target Audience & Problem Statement

Audience 1: Government Housing Analysts / Policy Makers

- Require insights into:
 - Affordability
 - Regional inequality
 - Market pressure
 - Supply constraints

Audience 2: Real Estate Investors / Developers

- Need insights into:
 - High-growth regions
 - Stable vs volatile markets
 - Investment timing
 - Demand hotspots

Problem Statement

The Irish housing market has experienced significant changes driven by economic growth, population increases, and constrained housing supply. However, understanding how prices, sales volumes, and affordability vary across time, location, and property condition remains complex. This project aims to analyse official property transaction data to uncover market trends, geographic disparities, and pricing structures, enabling evidence-based decision-making for policy formulation and investment planning.

6. Exploratory Data Analytics (EDA) – Key Insights

6.1 Price Distribution

- Median Price: **€281,938**
- Mean Price: **€368,481**
- Distribution is **right-skewed**
- Median better represents typical prices

6.2 Geographic Patterns

- **Most expensive:** Dublin, Wicklow, Kildare
- **Least expensive:** Leitrim, Longford, Roscommon
- Strong **urban–rural divide**

6.3 Sales Volume

- Dublin accounts for ~30% of all sales
- Rural counties have low transaction volumes

6.4 Temporal Trends

- **December** shows highest sales volume
- **January** is consistently lowest
- Prices peak in **Q4**

6.5 Property Condition

- ~82% of sales are **second-hand**
 - New builds command higher prices
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7. Dashboards Overview

7.1 Dashboard 1 – Market Overview

Target Audience

- Government Housing Analysts / Policy Makers

Purpose

This dashboard provides a **high-level overview of the Irish residential property market (2019–2024)**. It summarises overall pricing levels, sales activity, and long-term market trends, acting as the **starting point** for the analysis.

Information Presented

- **KPI Summary:** Average, median, minimum, maximum property prices and total sales
- **Monthly Average Price Trend:** Shows long-term price movements
- **Monthly Sales Volume Trend:** Highlights market activity and seasonality

- **Interactive Filters:** Year, County, Price Category, Property Condition

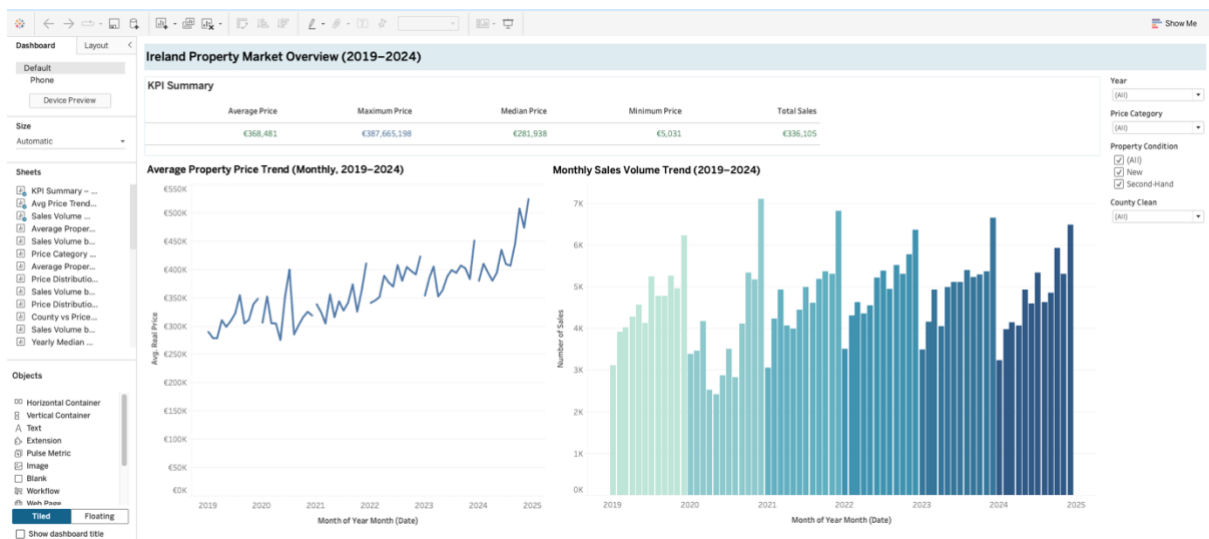
Key Insights

- Property prices show a **clear upward trend** over time
- Median price is lower than average, indicating a **right-skewed price distribution**
- Sales volumes fluctuate seasonally, with consistent overall demand

Actionable Insights & Decision Value

- **Policy makers** can identify affordability pressures and monitor market stability
- They can assess demand strength and timing of market activity

This dashboard supports **informed strategic decision-making** by providing a clear national-level market snapshot.



7.2 Dashboard 2 – County Market Overview

Target Audience

- Government Housing Analysts / Policy Makers

Purpose

This dashboard compares **property prices and sales volumes across counties**, focusing on the **top 15 counties by transaction volume**. It highlights **regional disparities** in pricing and market activity.

Information Presented

- **Average Property Price by County** (Top 15 by sales)
- **Sales Volume by County** (ranking of most active counties)
- **Price Category Distribution** (Low, Medium, High) by county
- **Interactive Filters:** County, Year, Property Condition, Price Category

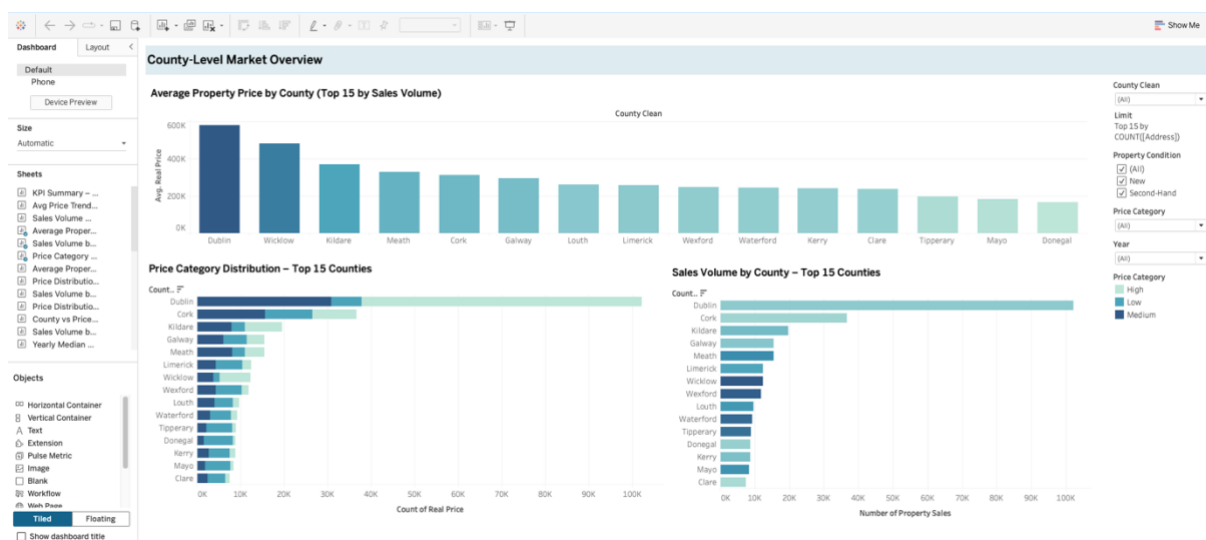
Key Insights

- Dublin has both the **highest average prices** and **highest sales volume**
- Counties surrounding Dublin (Wicklow, Kildare, Meath) show **elevated prices**
- Rural counties have **lower prices and lower transaction volumes**
- High-price category sales are concentrated in urban counties

Actionable Insights & Decision Value

- **Policy makers** can identify regions under affordability pressure and can compare high-demand vs lower-cost counties

This dashboard supports **regional planning and investment location decisions** by clearly visualising county-level market differences.



7.3 Dashboard 3 – Property Condition Analysis

Target Audience

- Government Housing Analysts / Policy Makers

Purpose

This dashboard analyses **price and sales differences between new and second-hand properties**. It highlights how property condition affects **pricing, distribution, and market supply**.

Information Presented

- **Average Property Price:** New vs Second-Hand
- **Sales Volume:** New vs Second-Hand
- **Price Distribution (Box Plot):** Comparison of price spread and outliers
- **Interactive Filters:** Year, County, Price Category, Property Condition

Key Insights

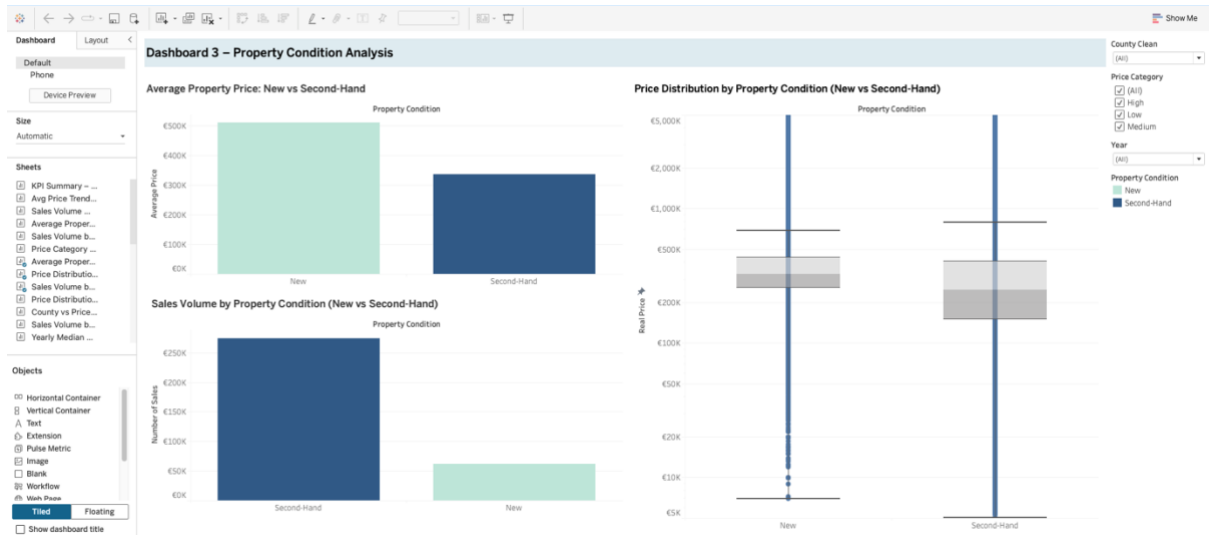
- New properties have a **higher average price** than second-hand properties

- Second-hand homes dominate **overall sales volume**, indicating supply constraints in new builds
- Price variability and outliers are present in both categories

Actionable Insights & Decision Value

- **Policy makers** can assess new housing supply shortages and affordability gaps
- Can evaluate demand and pricing premiums for new builds

This dashboard supports **decisions on housing supply planning and investment strategy**.



7.4 Dashboard 4 – Price Category Analysis

Target Audience

- Government Housing Analysts / Policy Makers

Purpose

This dashboard examines the **structure of the housing market using price categories (Low, Medium, High)**. It highlights how sales volumes and prices vary across counties and price segments.

Information Presented

- Sales Volume by Price Category
- Average Price by Price Category
- County vs Price Category Heatmap
- Interactive Filters: **Year, County, Property Condition, Price Category**

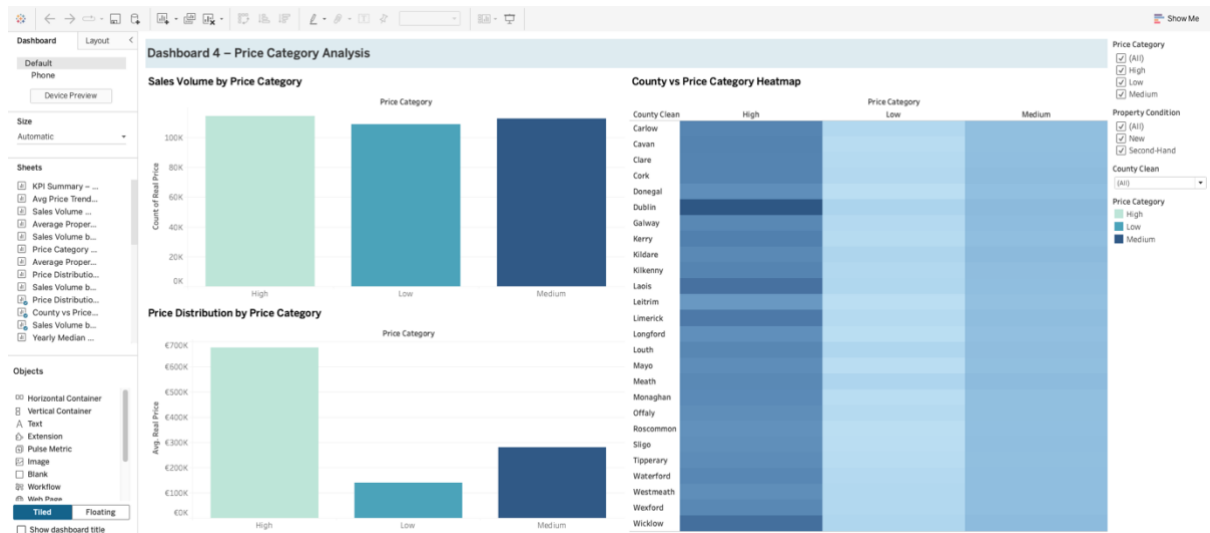
Key Insights

- Medium and High price categories account for a **large share of transactions**
- High-priced properties are concentrated in **urban and commuter-belt counties**
- Low-priced properties dominate **rural counties**

Actionable Insights & Decision Value

- **Policy makers** can identify affordability gaps and regional inequality
- Can target counties and price bands aligned with their strategy

This dashboard supports **price segmentation analysis and regional decision-making**.



7.5 Dashboard 5 – Market Trends Over Time

Target Audience

- Government Housing Analysts / Policy Makers

Purpose

This dashboard analyses **long-term market trends** by examining **yearly price movements, growth rates, and sales volumes**.

It focuses on how the housing market has evolved over time.

Information Presented

- Yearly Median Property Price Trend
- Year-on-Year (YoY) Median Price Growth %
- Yearly Sales Volume Trend
- Interactive Filters: **Year, County, Property Condition**

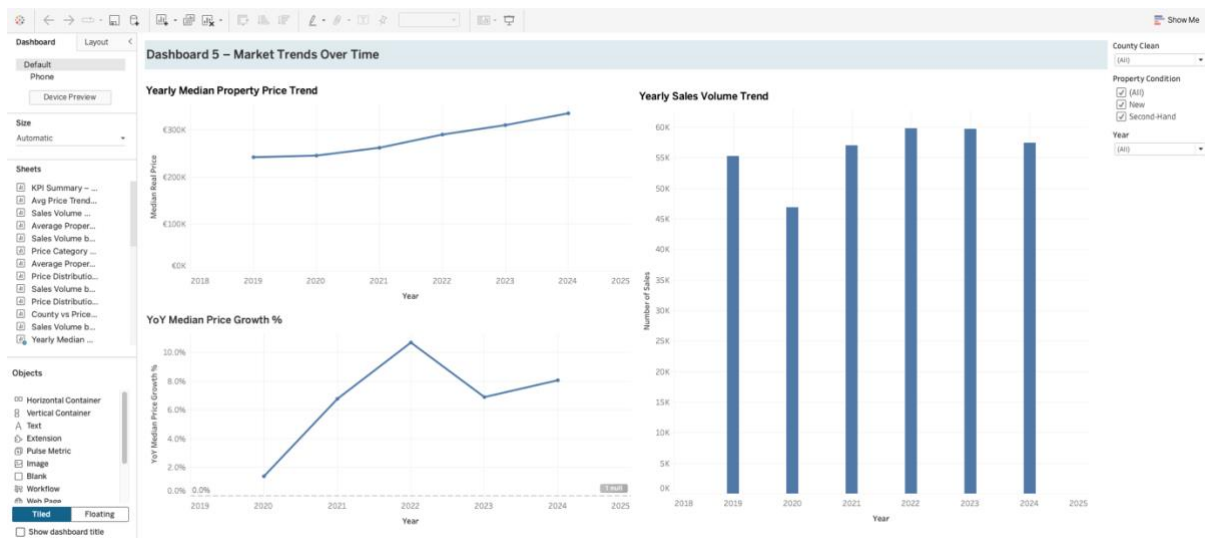
Key Insights

- Median property prices show a **consistent upward trend**
- YoY growth peaked around **2022**, followed by stabilisation
- Sales volumes remain strong despite rising prices

Actionable Insights & Decision Value

- **Policy makers** can monitor market growth and potential overheating
- Can assess market cycles and long-term price momentum

This dashboard supports **strategic, time-based decision-making**.



7.6 Dashboard 6 – Property Price Distribution & Spatial Analysis

Target Audience

- Real Estate Investors & Developers

Purpose

This dashboard explores **price distribution and geographic patterns** in the Irish housing market. It combines **statistical distribution analysis** with **spatial visualisation**.

Information Presented

- Property Price Distribution (Histogram)
- Median Property Price by County (Geo Bubble Map)
- Median Price vs Sales Volume by County (Scatter Plot)
- Interactive Filters: **Year, Property Condition**

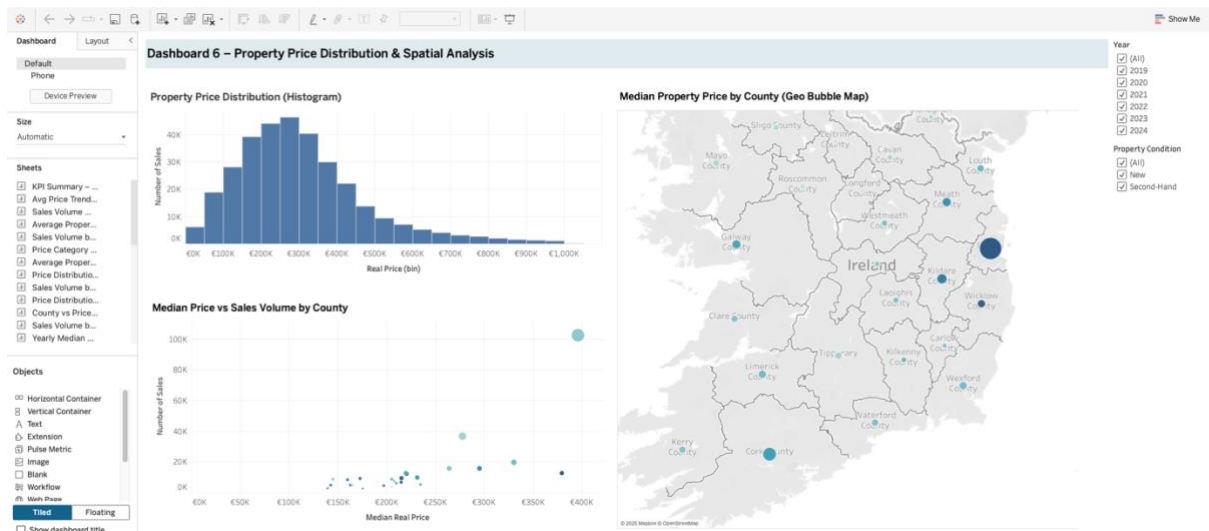
Key Insights

- Property prices are **right-skewed**, with most sales in the mid-price range
- Dublin and surrounding counties show **higher median prices**
- Higher-priced counties generally have **higher sales volumes**

Actionable Insights & Decision Value

- **Investors** can locate high-value and high-demand counties

This dashboard supports **spatial planning and location-based decision-making**.



7.7 Dashboard 7 – Sales Dynamics & Contribution Analysis

Target Audience

- Real Estate Investors & Developers

Purpose

This dashboard analyses **how different counties and price categories contribute to overall sales**, and how **monthly sales patterns vary across counties**.

Information Presented

- Sales Contribution by County and Price Category (Treemap)
- Monthly Sales Volume by County (Trellis View)
- Interactive Filters: **Year, County, Property Condition**

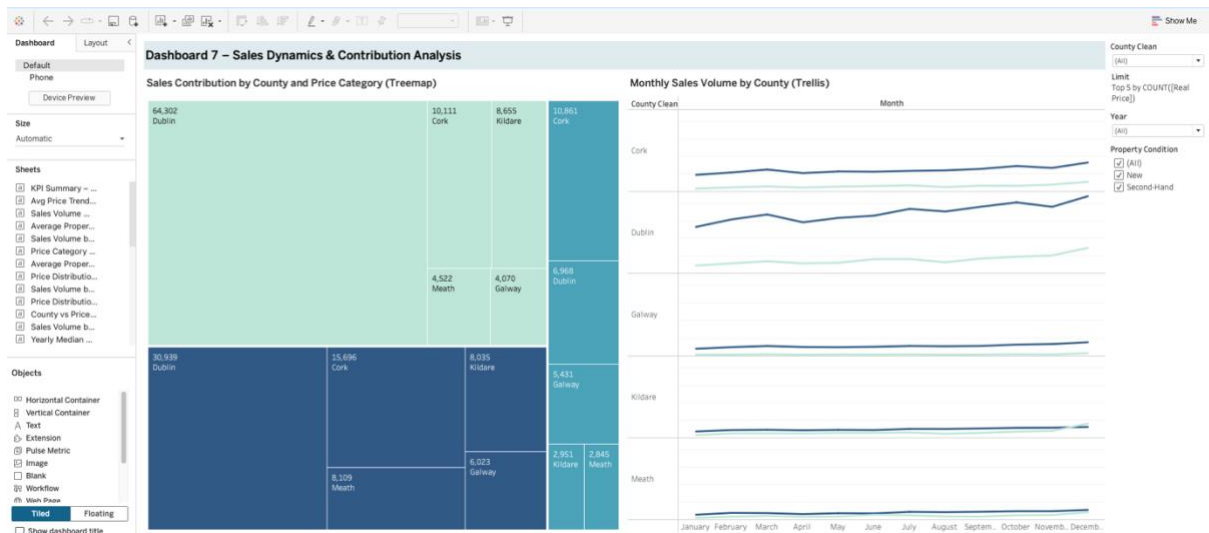
Key Insights

- Dublin contributes the **largest share of national sales** across price categories
- Cork, Kildare, and Meath are significant secondary contributors
- Monthly sales patterns show **consistent demand with seasonal variation**

Actionable Insights & Decision Value

- **Investors** can focus on high-contribution counties and stable markets

This dashboard supports **resource allocation and regional market prioritisation**.



7.8 Dashboard 8 – Quarterly Market Performance

Target Audience

- Real Estate Investors & Developers

Purpose

This dashboard evaluates **quarterly market performance** by analysing **price trends and sales volumes** across quarters.

It highlights seasonal patterns and differences by property condition.

Information Presented

- Quarterly Median Property Price Trend
- Quarterly Sales Volume by Property Condition (New vs Second-Hand)
- Interactive Filters: **Year, County, Property Condition**

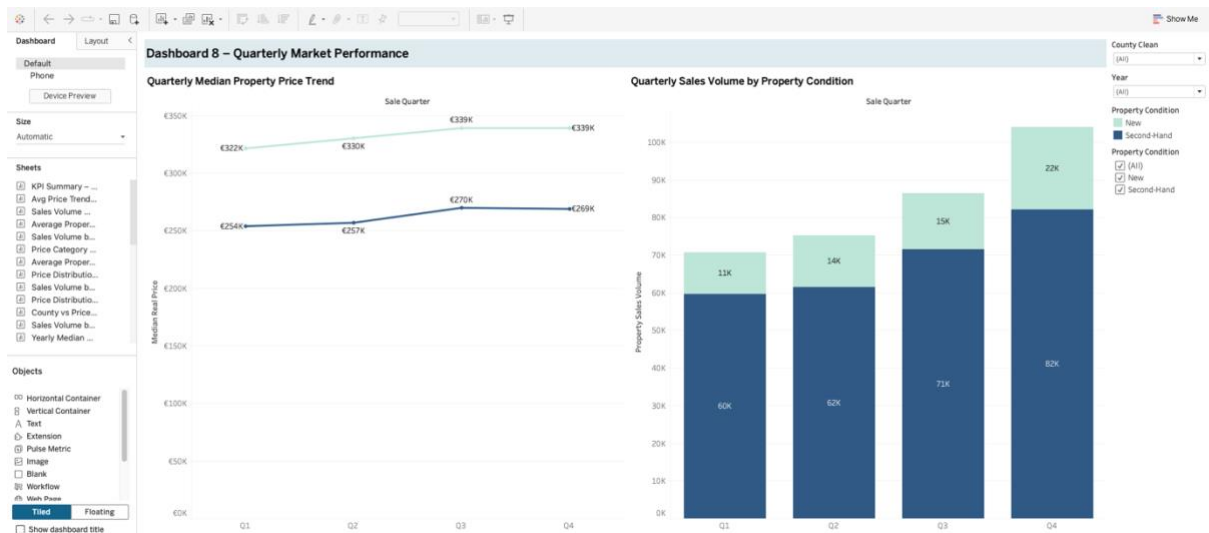
Key Insights

- Median property prices generally **increase from Q1 to Q4**
- Sales volumes peak in **Q4**, indicating strong year-end activity
- Second-hand properties dominate sales across all quarters

Actionable Insights & Decision Value

- **Investors and developers** can optimise transaction timing and project launches

This dashboard supports **quarter-based planning and forecasting**.



7.9 Dashboard 9 – Housing Affordability & Market Pressure by County

Target Audience

- Real Estate Investors & Developers

Purpose

This dashboard assesses **housing affordability and market pressure at county level** by combining **sales intensity and price volatility** indicators.

Information Presented

- Sales Pressure vs Affordability (Scatter Plot by County)
- Price Volatility by County (Box Plot / Index)
- Reference Lines: **National median and average benchmarks**
- Interactive Filters: **Year, Property Condition**

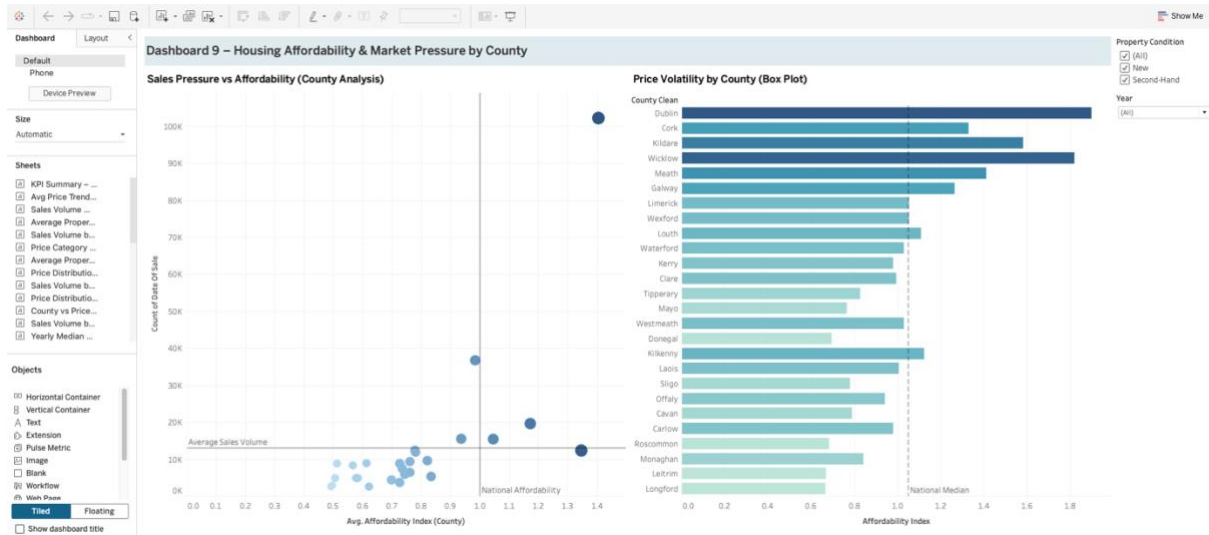
Key Insights

- Counties with high sales pressure also show **lower affordability**
- Urban counties experience **higher price volatility**
- Rural counties are generally **more affordable but less active**

Actionable Insights & Decision Value

- **Investors** can assess risk by comparing volatility and demand

This dashboard supports **targeted housing policy and risk-aware investment decisions**.



7.10 Dashboard 10 – Market Composition & Sales Intensity

Target Audience

- Real Estate Investors & Developers

Purpose

This dashboard provides insights into **market composition and sales intensity** by analysing **property condition shares** and **county-level sales activity across price categories**.

Information Presented

- Sales Composition by Property Condition (Pie Chart)
- County vs Price Category – Sales Intensity Heat Map
- Interactive Filters: **Year, County, Property Condition**

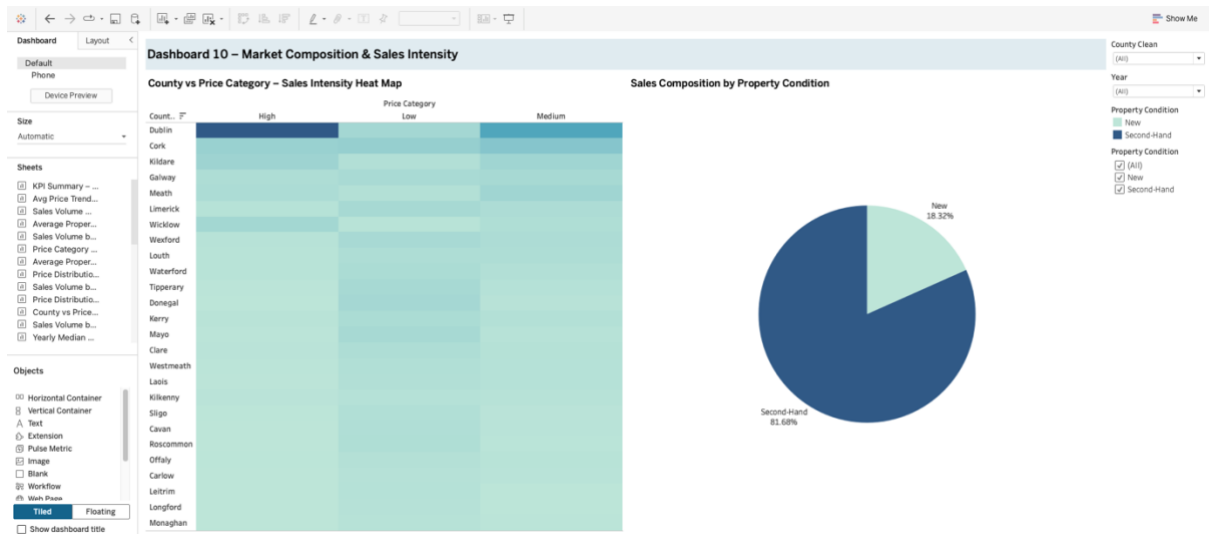
Key Insights

- Second-hand properties account for the **majority of sales (~82%)**
- High-price sales intensity is concentrated in **Dublin and surrounding counties**
- Rural counties show more balanced or low-intensity sales across price categories

Actionable Insights & Decision Value

- **Investors and developers** can assess dominant market segments and regional demand

This dashboard supports **market composition analysis and supply-side decision-making**.



8. Conclusion:

This exploratory analytics phase successfully:

- Identified major market patterns
- Highlighted affordability issues
- Revealed geographic inequalities
- Established strong analytical foundations

The dashboards provide **clear, interactive, and actionable insights** that directly support:

- Housing policy formulation
- Investment strategy
- Market monitoring